

OPERATORS QUESTION SHEET

Logical Operator

1. `System.out.println(true || false && false);` // Output?
2. `System.out.println((5 > 3) && !(2 < 1 || 3 == 3));` // Output?
3. `boolean a = false, b = true;`
`System.out.println(a && (b = false));` // Output? What is value of b?
4. `System.out.println(!(true && false || true));` // Output?
5. `System.out.println((10 > 5) || (5 / 0 == 1));` // Output?
6. `System.out.println((false && true) || (true && true));` // Output?
7. `System.out.println((10 < 5) || (20 > 10 && 5 < 2));` // Output?
8. `boolean x = true, y = false, z = true;`
`System.out.println((x || y) && !(y && z));` // Output?
9. `System.out.println(!(5 == 5 && 10 > 2) || (false));` // Output?
10. `System.out.println((true && false) || (false || true));` // Output?

Arithmetic Operator

11. `System.out.println(10 + 2 * 5 - 3);` // Output?
12. `System.out.println((10 + 2) * (5 - 3));` // Output?
13. `System.out.println(20 / 5 + 3 * 2 - 1);` // Output?
14. `System.out.println(100 % 3 * 4 + 6 / 2);` // Output?
15. `System.out.println(10 + 20 / 5 * 2);` // Output?
16. `System.out.println(5 + 2 * 3 % 4);` // Output?
17. `System.out.println((12 - 4) * 2 + 3);` // Output?
18. `System.out.println(15 / 2 * 2);` // Output? (int division)
19. `int a = 5;`
`System.out.println(++a + a++);` // Output?
20. `int x = 10;`
`System.out.println(x++ + ++x + x);` // Output?

Relational Operator

```
21. System.out.println((10 > 5) == (5 < 10));           // Output?
22. System.out.println(5 >= 5 && 6 != 7);               // Output?
23. System.out.println('A' == 65);                     // Output?
24. System.out.println('B' > 'A' && 'a' > 'Z');          // Output?
25. System.out.println(5 < 3 || 2 == 2 && 3 > 1);        // Output?
26. System.out.println((10 == 10) && !(10 != 10));      // Output?
27. int a = 10, b = 20, c = 30;
    System.out.println(a < b && b < c);                 // Output?
28. System.out.println((5 < 6) == true);               // Output?
29. System.out.println((true == false) != (false == false)); // Output?
30. System.out.println((5 < 10 ? 1 : 0) == 1);         // Output?
```

Complex Increment/Decrement Output Questions

```
1. int a = 5;
   System.out.println(++a + a++);                     // Output?

2. int a = 10;
   int b = a++ + ++a + a;
   System.out.println(b);                             // Output?

3. int x = 3;
   x = x++ + ++x;
   System.out.println(x);                             // Output?

4. int i = 1;
   int result = i++ + i++ + i;
   System.out.println(result);                         // Output?

5. int a = 2, b = 3;
   int c = ++a * b-- + a--;
   System.out.println(c);                             // Output?

6. int a = 5;
   int b = a++ + ++a + a++ + --a;
   System.out.println("a = " + a + ", b = " + b);    // Output?
```

7. `int a = 1;`
`a = a++ - --a + ++a - a--;`
`System.out.println(a);` // Output?
8. `int x = 4, y = 5;`
`int z = ++x + y-- - x++ + --y;`
`System.out.println(z);` // Output?
9. `int x = 6;`
`System.out.println(x++ + x++ + ++x);` // Output?
10. `int a = 10;`
`a = a++ + ++a - --a + a--;`
`System.out.println(a);` // Output?
11. `int a = 2;`
`int b = a++ + ++a + a + --a + a--;`
`System.out.println("a = " + a + ", b = " + b);` // Output?
12. `int a = 1, b = 1;`
`int c = a++ + b++ + ++a + ++b;`
`System.out.println(c);` // Output?
13. `int a = 0;`
`int b = a++ + a++ + a++ + a++;`
`System.out.println(b);` // Output?
14. `int a = 1;`
`int b = a++ + ++a + a++ + ++a;`
`System.out.println("a = " + a + ", b = " + b);` // Output?
15. `int x = 3;`
`int y = x++ + ++x + x++ + --x;`
`System.out.println(y);` // Output?
16. `int x = 1;`
`x = x++ + x++ + ++x + --x;`
`System.out.println(x);` // Output?
17. `int a = 2;`
`int b = --a + a++ + ++a + a--;`
`System.out.println(b);` // Output?

18. `int x = 5;`
`int y = ++x + x++ + --x + x--;`
`System.out.println("x = " + x + ", y = " + y); // Output?`
19. `int a = 3, b = 4;`
`int c = a++ + b++ + ++a + --b + a + b;`
`System.out.println(c); // Output?`
20. `int a = 1;`
`int b = a++ + ++a * a-- + --a;`
`System.out.println(b); // Output?`
21. `int a = 1;`
`a = a++ + ++a * (a++ + --a) - a--;`
`System.out.println(a); // Output?`
22. `int x = 2, y = 3;`
`int z = x++ * --y + ++x * y-- + x++ - y;`
`System.out.println("x = " + x + ", y = " + y + ", z = " + z); // Output?`
23. `int a = 5;`
`int b = a++ + a++ + ++a + a-- + --a;`
`System.out.println("a = " + a + ", b = " + b); // Output?`
24. `int a = 1;`
`int b = a++ + (a++ * (a++ + --a) + a--);`
`System.out.println("a = " + a + ", b = " + b); // Output?`
25. `int i = 0;`
`i = i++ - --i + ++i + i-- - i++ + ++i;`
`System.out.println(i); // Output?`
26. `int a = 1, b = 2, c = 3, d = 4;`
`int result = a++ + ++b * c -----d;`
`System.out.println("a=" + a + ", b=" + b + ", c=" + c + ", d=" + d + ", result=" + result);`

27.

```
int a = 2, b = 3, c = 4, d = 5;  
int result = --a + b++ - c-- + ++d + a++ * --b;  
System.out.println("a=" + a + ", b=" + b + ", c=" + c + ", d=" + d + ", result=" +  
result);
```

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```
int a = 5, b = 4, c = 3, d = 2;  
int result = a-- + ++b * --c - d++ + b-- + c;  
System.out.println("a=" + a + ", b=" + b + ", c=" + c + ", d=" + d + ", result=" +  
result);
```

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```
int a = 3, b = 2, c = 1, d = 0;  
int result = ++a * b-- + c++ - --d + d++ * --b;  
System.out.println("a=" + a + ", b=" + b + ", c=" + c + ", d=" + d + ", result=" +  
result);
```

30.

```
int a = 1, b = 2, c = 3, d = 4;  
int result = a++ + b++ + c++ + d++ + --a + --b + --c + --d;  
System.out.println("a=" + a + ", b=" + b + ", c=" + c + ", d=" + d + ", result=" +  
result);
```